

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI architecture that combines information retrieval with large language models (LLMs). Traditional LLMs generate responses based on knowledge learned during training, which can become outdated or incomplete. RAG addresses this limitation by retrieving relevant information from external sources before generating an answer. The retrieved documents are provided to the language model as context, enabling more accurate, reliable, and domain-specific responses. RAG is widely used in document question answering, enterprise knowledge systems, chatbots, and search applications.

A typical RAG workflow starts with collecting documents from sources such as PDFs, websites, databases, or text files. These documents are then processed and prepared for efficient retrieval. By combining retrieval and generation, RAG reduces hallucinations and improves the factual accuracy of AI-generated content.

Components of a RAG Pipeline

A RAG system typically consists of several stages. First, documents are loaded using document loaders. The text is then split into smaller chunks using text splitters. Chunking ensures that relevant information can be retrieved efficiently without exceeding the context window of the language model. Next, embeddings are generated using embedding models such as Sentence Transformers. Embeddings convert text into numerical vectors that capture semantic meaning.

The embeddings are stored in a vector database such as FAISS, ChromaDB, Pinecone, or Weaviate. When a user submits a query, the query is converted into an embedding and compared with stored document embeddings. The most relevant chunks are retrieved and passed to the language model. The LLM then generates an answer using both the query and retrieved context.

Advantages and Applications of RAG

RAG offers several advantages over standalone language models. It improves factual accuracy, allows access to up-to-date information, reduces hallucinations, and enables domain-specific knowledge integration. Organizations can build custom chatbots that answer questions from internal documents without retraining large models. This significantly reduces cost and development time.

Common applications include customer support systems, legal document analysis, healthcare knowledge assistants, educational platforms, enterprise search engines, and research assistants. Modern RAG systems often use frameworks such as LangChain and LlamaIndex, vector databases like FAISS, and LLMs from providers such as OpenAI, Groq, Anthropic, and Google. As AI adoption grows, RAG has become one of the most important techniques for building practical and trustworthy AI applications.